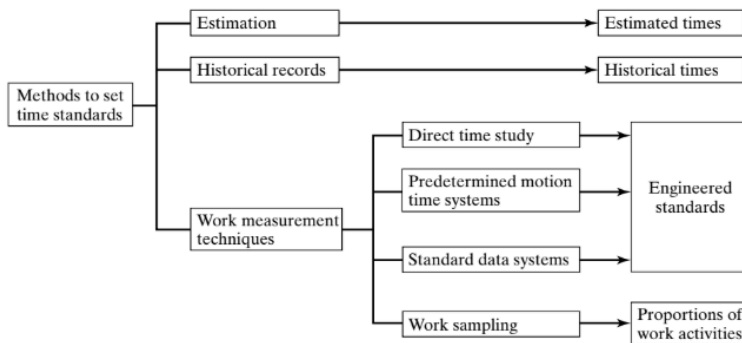


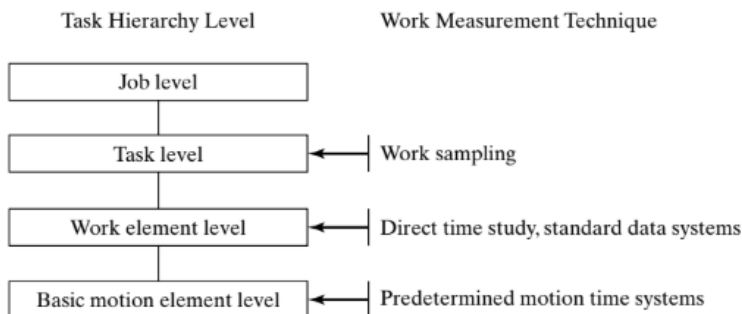
→ The objective of these work measurement techniques is to determine a standard time for the task.
→ Time study refers to all of the ways in which time is investigated and analyzed in work situations, whether the work is accomplished by human workers or automated systems.
→ Time standards are required where :

- Low Productivity
- Repetitive cycles
- Short cycle times
-

Methods to Determine Time Standards



Work Measurement Techniques



→ Direct Time Study

→ Predetermined Motion Time Systems–A *predetermined motion time system (PMTS)* relies on a database of basic motion elements (*therbligs*) such as reach, grasp, and moves that are common to nearly all manual industrial tasks.

There are two advantages related to predetermined motion time systems:

(1) performance rating is not required

(2) they can be applied to determine the time standard for a task before production.

→ Standard Data Systems

→ Work Sampling

PREREQUISITES FOR VALID TIME STANDARDS

- The task is performed by an average worker
- The worker's pace represents standard performance

- The worker uses the standard method
- The task is performed on a standard work unit that is defined before and after processing.

→ ***The standard method is the procedure that has been determined to be the optimum method for processing a work unit.***

- Procedure (actions and motions) used by the worker.
- Tools.
- Equipment
- Workplace layout.
- Irregular work elements
- Working conditions
- Setup

ALLOWANCES IN TIME STANDARDS

- (1) scheduled break periods during the shift and
- (2) a PFD allowance added to the normal time.

- Scheduled Break Periods.
- The PFD Allowance.
- Contingency Allowances. The contingencies for which these allowances are provided are usually because of some kind of problem with the task or the production equipment used to perform it.
- Policy Allowances. These allowances are intended to cover special work situations that are usually associated with a wage incentive system.

→ Accuracy is the degree to which the measured value agrees with the true value of the quantity of interest.

→ Precision refers to the repeatability of the measurement system.

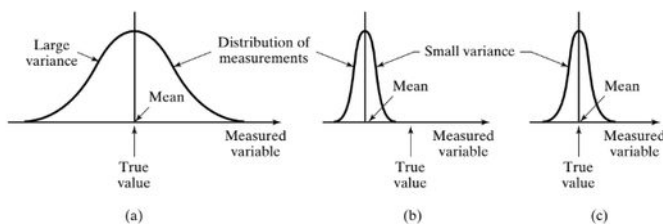


Figure 6 Accuracy versus precision in measurement: (a) high accuracy but low precision, (b) low accuracy but high precision, and (c) high accuracy and high precision.

Methods of determining time standards ranked by relative accuracy

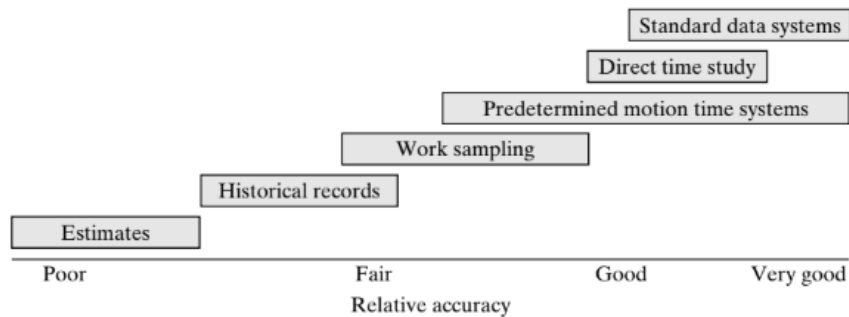


Figure 7 Methods of determining time standards ranked by relative accuracy.

Methods of determining time standards ranked by relative application Speed

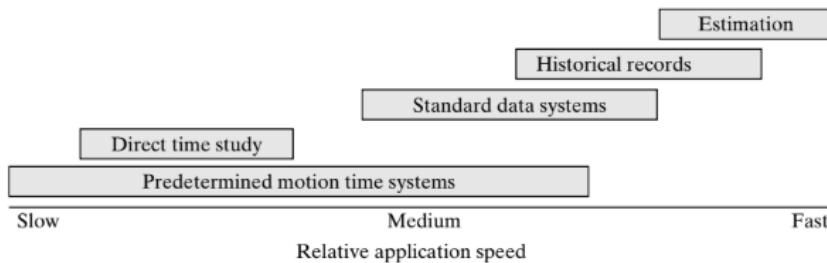


Figure 8 Methods of determining time standards ranked by relative application speed.

Direct time study

→ Direct time study, also known as stopwatch time study

The procedure for determining the standard time for a task using direct time study can be summarized in the following five steps:

1. **Define and document the standard method.** - the “one best method” principle
2. **Divide the task into work elements.** - work element is a series of motion activities that are logically grouped together because they have a unified purpose in the task.
3. **Time the work elements** to obtain the observed time for the task.
(1) snapback timing method and (2) continuous timing method.

In the **snapback timing method**, the watch is started at the beginning of every work element by snapping it back to zero at the end of the previous element.

In the **continuous timing method**, the watch is zeroed at the beginning of the first cycle and allowed to run continuously throughout the duration of the study.

4. **Evaluate the worker's pace relative to standard performance**, a procedure called performance rating. This is used to determine the normal time. Steps 3 and 4 are accomplished simultaneously.
5. Apply an **allowance to the normal time** to compute the standard time.

Predetermined Motion Time Systems

A predetermined motion time system (PMTS), also called a predetermined time system, is a database of basic motion elements and their associated normal time values, together with a set of procedures for applying the data to analyze manual tasks and establish standard times for the tasks.

The PMTS Procedure

To apply a predetermined motion time system to a task, the following steps:

1. Synthesize the method that would be used by a worker to perform the task (or analyze the method that is being used by a worker in an existing task).
2. Retrieve the normal time value for each motion element, based on the work variables and conditions under which the element will be (is) performed.
3. Evaluate the method to make improvements by eliminating motions, reducing distances, introducing special tools, using simultaneous right and left hand motions, and so on.
4. Apply allowances to the normal time to determine the standard time for the task.